

Classification of Ear Diseases from Otoscopic Images Using DenseNet121 and Grow-and-Learn Network

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Abstract—Diagnosis of ear diseases from otoscopy images is of critical importance to initiate the treatment process accurately and quickly. In this study, a 4-class classification problem is solved using Ear Imagery Database containing images labeled chronic otitis media, earwax plug, myringosclerosis and normal. InceptionV3, Xception, ResNet50, DenseNet121, VGG19 and MobileNet models were used to extract features from images and Grow-and-Learn (GAL) network was used for classification. In GAL network, the number of nodes is automatically adjusted and updated during the training process when needed. The network expands as it learns new information and shrinks as it starts to forget. GAL has a structure similar to KNN (K-Nearest Neighbor) algorithm in the classification phase, but it is a network that can perform higher performance classification with fewer nodes. In this way, it can be used as an alternative to KNN and the classification process can be performed in a shorter time. With this method, ear images were classified into different disease states and the performance of the proposed model was evaluated. Experimental results show that the best performance is achieved with DenseNet121+GAL with a maximum of 98.86% and an average of 97.73%. DenseNet121+KNN achieves a 96.21% success. Experiments show that GAL provides faster and more accurate classification using fewer nodes. This approach can help experts diagnose diseases quickly and accurately, thus providing more effective and timely intervention.

Index Terms—ear disease classification, deep learning, grow-and-learn network.

I. INTRODUCTION

Ear diseases are among the health problems that are common worldwide and negatively affect the quality of life of individuals [1]. These diseases occur with symptoms such as hearing loss, earache, ringing and balance problems. These disorders are more common especially in children, the elderly and individuals with weakened immune systems [2]. If left untreated, it can lead to permanent hearing loss, chronic infections or more serious complications [3]. Therefore, correct diagnosis is of great importance in terms of preserving the quality of life of patients and preventing progressive health problems.

The most commonly used methods in the diagnosis of ear diseases are otoscopic examination and audiometry tests

[4]. Otoscopic examination allows direct observation of the eardrum and external auditory canal, while audiometry allows the assessment of hearing ability [5]. These methods can provide reliable results when applied by experienced physicians. However, since the diagnostic process is largely based on the experience of the physician, different interpretations may occur in some cases. Evaluation of otoscopic images is subjective and may require additional tests. Incorrect or delayed diagnoses may lead to progression of the disease and delays in the treatment process [6]. This situation increases the need for new methods that will allow faster and more reliable diagnosis of ear diseases.

In recent years, artificial intelligence and image processing techniques have been increasingly used in medical diagnosis processes. In particular, automatic classification of diseases from otoscopic images can speed up the diagnosis process and make it more reliable [7]. In this study, the Grow and Learn (GAL) network [8] was used to make a more effective classification in the diagnosis of ear diseases. The features extracted with DenseNet121 were classified with the GAL network. GAL network is an artificial neural network model that can continuously optimize the network structure thanks to dynamic learning and forgetting mechanisms [9].

The contributions of this study can be listed as follows:

- To our knowledge, the GAL neural network was used for the first time in the diagnosis of external and middle ear diseases.
- Since the GAL network was used for classification purposes, classification was performed with fewer nodes and in a shorter time.
- The classification performance obtained is comparable to the literature.

II. LITERATURE REVIEW

There are many studies in the literature on the detection of ear diseases using deep learning and machine learning. The studies conducted with the dataset used in this research are summarized below.

Afify et al. [10] used convolutional neural networks (CNN) and Bayesian hyperparameter optimization for the diagnosis of ear diseases and achieved 98.10% accuracy.

Demircan et al. [11] compared the model performances in the classification of ear diseases using models such as AlexNet, ResNet50, ResNet101, ResNet50V2, ResNet101V2, InceptionV3, Xception and InceptionResNetV2. In the analysis, the highest accuracy was obtained by the InceptionResNetV2 model with 94%.

Viscaino et al. [12] developed a machine learning-based computer-aided diagnosis system. They evaluated the performances of three different classifiers, namely SVM, KNN and decision trees, using feature extraction methods such as color harmony vector, discrete cosine transform and filter bank. The highest accuracy was obtained with the SVM model with 93.9%.

Demircan et al. [13] proposes a model based on visual transformer (ViT) using attention mechanism. The features extracted with ViT are evaluated with KNN, SVM and random forest classifiers. The highest success is achieved by the ViT-KNN combination with 99% accuracy.

Akyol et al. [14] proposes a soft voting based ensemble learning method. They obtained 98.8% accuracy by combining the predictions of Inception V3, DenseNet121, VGG16, MobileNet and EfficientNetB0 models.

In this study, DenseNet121 was utilized for feature extraction and GAL network was used for classification. The tests resulted in a maximum accuracy of 98.86%, with precision, recall, and f-score metrics each achieving a value of 0.988.

III. MATERIALS AND METHODS

In this study, the publicly available Ear Imagery Database served as the dataset. DenseNet121 was employed to extract features from the images, while the GAL network was applied for classification purposes.

A. Ear Imagery Database

In this study, the publicly available Ear Imagery Database prepared by Viscaino and Cheein [15] was used to classify ear diseases. The dataset consists of 880 otoscopy images with a resolution of 420×380 belonging to 180 patients aged between 7 and 65 years who applied to Department of Otolaryngology of the Clinical Hospital of Universidad de Chile. Otoscopic images of both right and left ears are obtained from patients using a DE500 Firefly digital otoscope –with a 0.4 mm and 0.5 mm speculums– connected to a computer. The dataset contains images labeled as chronic otitis media, earwax plug, myringosclerosis and normal. Each image class contains 220 otoscopic images. Thus, the number of images in the classes is balanced. The images in the dataset included in this study were collected and labeled by a research team including an otolaryngologist. This process was carried out in collaboration with the Department of Ear, Nose and Throat of the Clinical Hospital of Universidad de Chile. Fig. 1 shows sample images belonging to these classes.

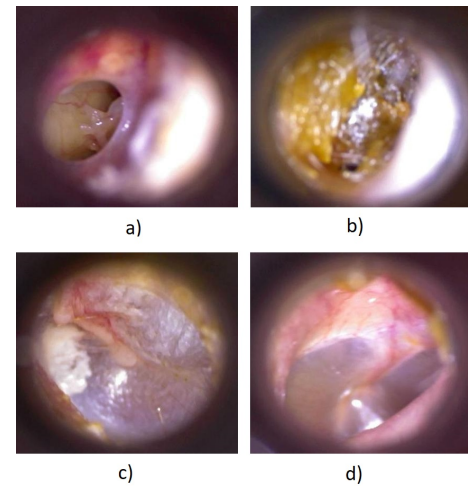


Fig. 1. Sample images from dataset a) chronic otitis media b) earwax plug c) myringosclerosis d) normal.

B. Method for Feature Extraction: DenseNet121

Densely connected CNNs (DenseNets) are a family of models that show high performance in deep learning tasks such as image classification [16]. The basic principle of this model is that each layer can directly access the feature maps from all previous layers, increasing parameter efficiency and improving gradient flow.

DenseNet121 is one of the widely used variants of this architecture, which has 121 layers. The model consists of four dense blocks, and each block contains fully connected convolutional layers. Since each layer takes the output of all previous layers as input, parameter sharing is more efficient and allows the model to generalize better by preventing unnecessary information repetition.

In addition, in the DenseNet121 model, the number of channels is kept under control and the computational cost is optimized by using transition layers. The model has yielded successful results on large-scale datasets such as ImageNet [17] and is widely used in deep learning-based image processing studies.

The advantages of DenseNet121 include reduced number of parameters, better information flow between deep layers, and more efficient training. Thanks to these features, it is preferred as an effective model, especially in the field of medical imaging with limited data [18].

C. Method for Classification: GAL Network

The GAL (Grow and Learn) network is an artificial neural network that can grow dynamically and automatically determine the number of nodes during the learning process. Its basic principle is that it performs classification based on the nearest distance rule. The first layer of this two-layer network determines the nearest node by calculating the minimum distance between the input vectors and the node weights and compares the class label. If the class of the input vector is not the same as the current node class, a new node is added

to the network. The second layer makes class decisions by combining the outputs of the nodes and brings together nodes belonging to the same class using the logical OR operation. The weights in this layer are initially 0 and are updated to 1 during the learning process. The configuration of the nodes in the network is defined by the following equations, and the structure of the GAL network is illustrated in Fig. 2.

$$D_j = \sum_{i=1}^n (x_i - w_{ji})^2 \quad E_e = \begin{cases} 1 & D_e = \min_j \{D_j\} \\ 0 & \text{Otherwise} \end{cases} \quad (1)$$

$$\text{Output layer: } C_c = \sum_e E_e \cdot T_{ec} \quad (2)$$

Here, x_i represents the i -th element of the input feature vector X . W_j refers to the weight vector associated with the exemplar node E_j , while w_{ji} represents the i -th element of W_j . D_j indicates the distance between the input vector X and W_j . Only one exemplar node, E_j , is active at a time, specifically the one whose weight vector is closest to the input vector, which in turn activates the corresponding class node. C represents the layer of class nodes. E_e represents the output of the nodes in the first layer, T_{ec} is the connection between an exemplar node e and a class node c , where the value of T_{ec} is either 1 or 0 depending on whether e is an exemplar of class c [9].

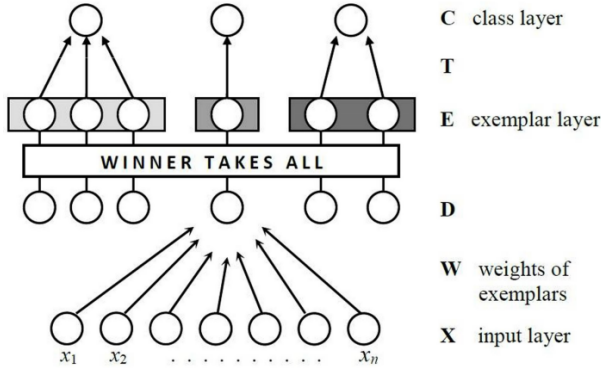


Fig. 2. The structure of GAL network.

The GAL network uses the nearest distance rule when determining class boundaries. In two-dimensional space, a line is formed between nodes from two classes and this line represents the class boundary, while in multidimensional space, a hyperplane is placed between nodes from different classes. The determination of the boundaries between classes depends on the training data and the distribution of nodes. Two processes are applied in the creation of the GAL network: the learning process and the forgetting process. In the learning process, a random vector from each class is selected from the training set and the network is initially created with these nodes. Then, a random vector is taken from the training set and the distances to the existing nodes are calculated. If the class of the closest node is the same as the class of the input vector, a new input vector is taken by decreasing the iteration number and the process is repeated. If the classes are different, the

input vector is added as a new node and this process continues until the specified iteration is completed.

In the GAL network, nodes that were previously meaningful but later became unnecessary are cleaned with the forgetting algorithm. In this process, nodes belonging to the network are temporarily removed in order and the removed node is presented again as the input vector. If the network can classify correctly without this node, the node is permanently removed. This process continues until the iteration number reaches zero and enables the network to work more efficiently by cleaning unnecessary nodes. Algorithm 1 shows the pseudocode for the learning and forgetting processes of the GAL network.

The biggest advantage of the GAL network is that it has a fast training process. The number of nodes is determined automatically and unnecessary nodes are removed. It does not require parameter optimization and can adapt to the learning data. However, it also has some disadvantages. The first of these is that the network has a rote training model, because all input vectors are assigned to the nodes one-to-one. Also, if the class boundaries are too complex, there is a risk of generating excessive nodes. If the feature selection is poor, the performance of the network may deteriorate.

Algorithm 1 GAL Network Learning and Forgetting Algorithm

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1: Initialize the network with one node for each class
2: Set ITERATION_COUNT to a predefined value
3: while ITERATION_COUNT > 0 do
4:   Select a random input vector  $X$  from the training set
5:   Find the closest node  $D_{min}$  to  $X$ 
6:   if  $\text{Class}(D_{min}) == \text{Class}(X)$  then
7:     Decrease ITERATION_COUNT
8:     Continue to next input
9:   else
10:    Create a new node  $D_{new}$  with weights equal to  $X$ 
11:    Set its corresponding second-layer connection to 1
12:   end if
13: end while
14: Print "Learning Process Completed"
15: ITERATION_COUNT = Number of nodes in the network
16: while ITERATION_COUNT > 0 do
17:   Temporarily remove a node  $D_k$  from the network
18:   Use  $D_k$  as an input and find the closest node  $D_{min}$  to  $D_k$ 
19:   if  $\text{Class}(D_{min}) == \text{Class}(D_k)$  then
20:     Permanently remove  $D_k$ 
21:   else
22:     Restore  $D_k$  to the network
23:   end if
24:   Decrease ITERATION_COUNT
25: end while
26: Print "Forgetting Process Completed"

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Fig. 3 shows the general scheme of the classifier used in this study.

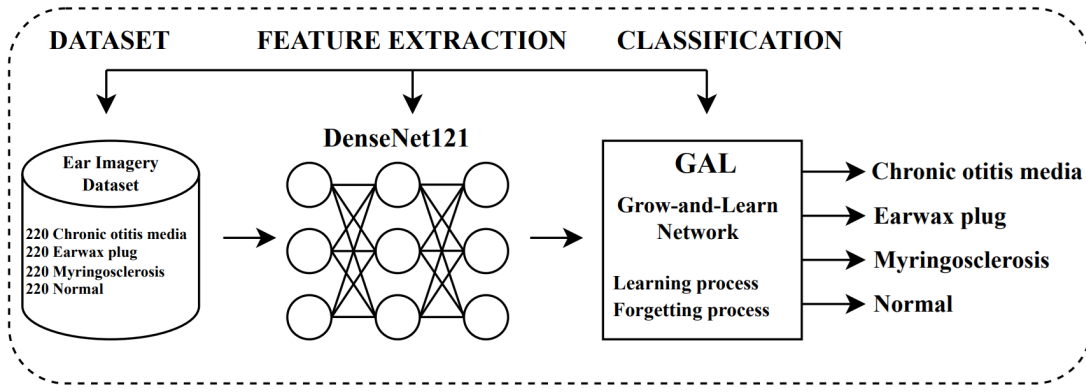


Fig. 3. The general scheme of the ear disease classifier.

IV. EXPERIMENTAL RESULTS

The tests were conducted on a system featuring a 14th Gen Intel Core™ i7-14700 vPro processor (2.1 GHz base, 5.4 GHz Turbo), an NVIDIA GeForce RTX 4060 graphics card (8 GB GDDR6X), and 32 GB of DDR5 RAM.

A. Performance Metrics

In this study, accuracy, precision, recall and f-score metrics were used to evaluate the classification performance of the models used. The equations for these metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F-score} = \frac{TP}{TP + 0.5(FP + FN)} \quad (6)$$

Accuracy is the ratio of correct predictions to the total number of observations. Here TP is true positives, TN is true negatives, FP is false positives, and FN is false negatives. Precision shows how many of the model's positive predictions are correct. Recall shows how many of the true positives are correctly predicted. F-score is the harmonic mean of Precision and Recall and tries to balance both metrics.

B. Test Results

In this study, ear diseases were classified using Ear Imagery Database. Images were divided into two groups as 70% training set and 30% test set. Feature extraction was performed using models that are commonly used in the literature and have demonstrated success. The GAL network was employed for classification. In order to show the effectiveness of GAL network, images were first classified with KNN. K value was used as 3 in KNN. Then, classification was performed using GAL network. The results of the tests are shown in Table I and Table II.

TABLE I
CLASSIFICATION ACCURACIES OF DEEP LEARNING MODELS WITH KNN

Models	Accuracy (%)
InceptionV3+KNN	93.18
Xception + KNN	95.08
ResNet50 + KNN	97.73
DenseNet121 + KNN	96.21
VGG19 + KNN	95.08
MobileNet + KNN	96.97

According to Table I, the highest performance of 97.73% is observed, which was achieved using the features obtained from the ResNet50 model. In the experiments, 616 samples were used in the training set and 264 in the test set. This means that each sample in the test set was compared with each sample in the training set to determine its class, and the class of the test sample was determined according to the number of K neighborhoods by calculating the shortest distances.

Upon reviewing Table II, it can be seen that five tests were conducted for each model to assess their performance, and the results are provided as averages. The best classification result was achieved by using the features extracted with DenseNet121 with a maximum of 98.86% and an average of 97.73% success. In addition, it is seen that the GAL network consists of 102 nodes in the model where the best performance was obtained. This means that in order to determine the class of each sample in the test set, it was compared with each of the 102 nodes present in the GAL network and its class was determined according to the nearest distance rule. While the network to which the test samples will be compared in KNN consists of 616 nodes, it consists of 102 nodes in the GAL network. We can also see from Table II that the ResNet50+GAL classifier showed a good performance of 98.48% with only 90 nodes present in the network.

Fig. 4 presents the confusion matrix for the test that achieved the highest classification performance. The accuracy, precision, recall and f-score values for this test are obtained as 98.86%, 0.988, 0.988 and 0.988, respectively. The obtained

TABLE II
PERFORMANCE COMPARISON OF DEEP LEARNING MODELS WITH GAL

Models Tests	InceptionV3+GAL		Xception + GAL		ResNet50 + GAL		DenseNet121 + GAL		VGG19 + GAL		MobileNet + GAL	
	Accuracy	#Node	Accuracy	#Node	Accuracy	#Node	Accuracy	#Node	Accuracy	#Node	Accuracy	#Node
1	92.80	128	95.83	117	94.32	93	97.72	99	95.45	127	96.97	112
2	93.94	122	94.32	116	97.35	88	96.97	95	97.73	125	96.21	100
3	92.42	120	94.32	101	97.73	88	98.86	102	96.97	137	97.35	113
4	95.08	132	94.32	108	98.48	90	96.97	92	95.45	130	95.45	104
5	94.32	127	97.35	109	98.11	92	98.11	99	96.59	126	97.73	110
Average	93.71	126	95.23	110	97.20	90	97.73	97	96.44	129	96.74	108

TABLE III
PERFORMANS COMPARISON WITH OTHER STUDIES

Authors	Feature Extractor	Classifier	Acc. (%)
Afify et al. [10]	CNN	FCNN	98.10
Demircan et al. [11]	InceptionResNetV2	FCNN	94.0
Viscaino et al. [12]	Color Coherence Vec. Discrete Cos. Trans. Filter bank	SVM	93.9
Demircan et al. [13]	ViT	KNN	99.0
Akyol et al. [14]	Ensemble transfer learning models	FCNN	98.8
This study	DenseNet121	GAL	98.86

results are comparable to those in the literature. Table III shows a comparison of the proposed model with other studies in the literature that use the same dataset.

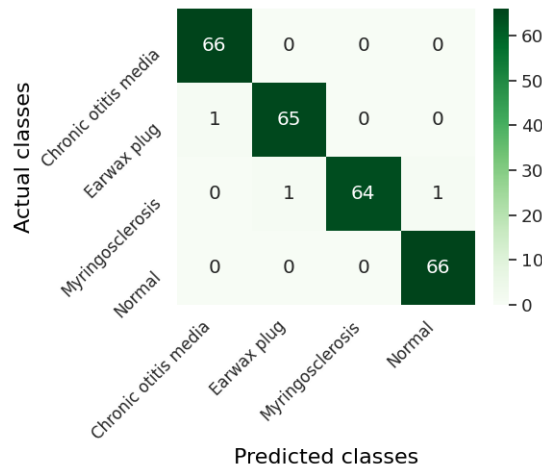


Fig. 4. Confusion matrix for the best model using Densenet121 + GAL.

V. CONCLUSIONS

In this study, ear diseases were detected from otoscopic images. DenseNet121 was used to extract features from images and GAL network was used to classify them; high accuracy rates were achieved. The experiments showed that GAL network can perform faster and more effective classification using fewer nodes compared to KNN thanks to dynamic node

management. In particular, DenseNet121+GAL combination exhibited the best performance with a maximum accuracy rate of %98.86.

One of the most important advantages of this study is that the classification was performed with classical machine learning methods. Since GAL network requires much fewer parameter calculations compared to fully connected neural networks (FCNN) used for classification in deep learning models, it offers a great advantage in terms of both processor usage and processing time. GAL network performs the classification process according to the nearest distance rule like KNN, but instead of comparing the test sample with all the samples in the training set, it compares it with fewer samples dynamically generated in the GAL network. Thanks to the dynamic optimization of the number of nodes, the classification process is completed in a much shorter time and system resources are used more efficiently. These features make the GAL network a powerful alternative in real-time applications and systems with low hardware requirements.

The results obtained show that the developed method can contribute to the rapid and accurate diagnosis of ear diseases and support the clinical decision-making processes of doctors. In the future, it is aimed to further improve the method with different datasets and additional deep learning models, and to make adaptations for practical use by testing in real clinical environments.

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